

HTML Lab Problems

Text & Formatting • Links & Images • Lists & Tables

Objective: Every problem must combine all three areas. Students should select the correct HTML element and attribute based on the information being represented.

Concepts to Demonstrate

- Text: h1–h6, p, br, hr.
- Formatting: b, strong, i, em, mark, u, pre, code, blockquote, sub, sup and other taught formatting elements.
- Links and images: a, img and their taught attributes, including clickable images.
- Lists: ol, ul, dl, nested lists and taught list attributes.
- Tables: table, tr, th, td, caption, rowspan, colspan and other taught table attributes.

Restriction: Do not use CSS or JavaScript. Use tables only for structured data and lists only when the information is actually a collection, sequence, or set of definitions.

Lab Problem 1: University Campus Guide

Scenario: A university wants a student-friendly campus guide covering the institution, facilities, student services, timings and useful resources.

Requirements

1. Use the university name as h1 and organize the page with a logical heading hierarchy.
2. Write detailed paragraphs for University Overview, Campus Life, Academic Facilities, Student Services and Contact.
3. Use strong, em, mark, b, i and another taught formatting element meaningfully.
4. Add a Principal/Director message using blockquote.
5. Include a small computer-lab example using pre and code.
6. Add at least four campus/facility images with meaningful src and alt values.
7. Make at least two images clickable and connect them to relevant information pages.
8. Create useful links for University Website, Admissions, Student Portal and Contact, using taught anchor attributes.
9. Create a ul for campus facilities, an ol for new-student orientation steps, and a dl for Library, Hostel, Laboratory, Auditorium and Student Portal.
10. Create a table with Facility, Location, Opening Time, Closing Time and Responsible Department, with at least six rows and a caption.
11. Use rowspan/colspan if a genuine relationship in the data requires merged cells.
12. Use hr between major sections and br for intentional address/contact line breaks.

Challenge: The page must feel like one real campus guide. Students must decide naturally which information belongs in prose, a list, a definition list, or a table.

Lab Problem 2: Travel Destination Explorer

Scenario: A travel website wants a destination page that helps visitors understand a city, discover attractions, compare experiences and find useful travel resources.

Requirements

1. Create a destination name as h1 with a strong introduction.
2. Create sections for About, Famous Attractions, Local Food, Culture, Best Time to Visit, Travel Tips and Resources.
3. Use multiple heading levels correctly.
4. Use strong for important advice, mark for the recommended season and em/i/u where appropriate.
5. Include a travel quote using blockquote.
6. Add at least five destination or attraction images with meaningful alt text.
7. Make at least two attraction images clickable and link them to relevant reference pages.
8. Add links to an official tourism site, an attraction page, a travel reference and a transport/resource page.
9. Use meaningful title and other taught anchor attributes.
10. Create a ul of recommended attractions, an ol for a one-day itinerary, and a dl defining Heritage Site, Local Guide, Public Transport, Peak Season and Off Season.

11. Create an attractions table with Attraction, Type, Best Time, Visit Duration and Reference Link; include at least six rows and a caption.
12. Use sub/sup meaningfully if appropriate and hr/br where justified.

Challenge: Students must distinguish an itinerary sequence from a general collection and make every image/link useful rather than decorative.

Lab Problem 3: Technology Product Showcase

Scenario: A technology company is launching a new laptop, smartphone, smartwatch or fictional device and wants a complete product showcase page.

Requirements

1. Create the product name as h1 with a tagline and detailed overview.
2. Create sections for Highlights, Features, Technical Specifications, Pricing, Comparison, Reviews and Support.
3. Use strong, em, mark, u and other taught formatting elements purposefully.
4. Display a technical example using pre and code.
5. Use sub/sup where meaningful for technical units or specifications.
6. Add at least four product images with meaningful alt text and taught image attributes.
7. Make at least three product images clickable, each leading to a relevant destination.
8. Add Buy Now, Full Specifications, Support and Company Website links with appropriate taught anchor attributes.
9. Create a ul of product highlights, an ol explaining how to choose a configuration, and a dl defining RAM, SSD, CPU, GPU, Refresh Rate and Warranty.
10. Create a comparison table containing Product, Processor, RAM, Storage, Display, Battery, Price and Warranty for at least four products.
11. Add a table caption and use rowspan/colspan only when logically necessary.
12. Add a short customer/expert statement using blockquote.

Challenge: The challenge is to combine marketing content, technical information, media, navigation, lists and a multi-column comparison table without CSS.

Lab Problem 4: College Event & Competition Portal

Scenario: A college is organizing an annual technology and cultural festival. Students need one page containing event information, competition details, schedule, rules and useful navigation.

Requirements

1. Create the event name as h1 and add an introductory description.
2. Create sections for About the Fest, Highlights, Schedule, Competitions, Rules, Prizes, Sponsors and Contact.
3. Use formatting elements to highlight deadlines, important rules and announcements.
4. Add an organizer message using blockquote.
5. Include a pre/code block showing a sample registration or participant reference format.
6. Add at least four event images with meaningful alt text.
7. Make at least two images clickable and connect them to registration or event information.
8. Add Registration, Official Website, Venue and Contact links using suitable anchor attributes.
9. Create a ul for general rules, an ol for participant registration steps, and a dl defining Individual Event, Team Event, Open Event, Coordinator, Preliminary Round and Final Round.
10. Create a three-day schedule table with Date, Time, Event, Venue, Category and Coordinator.
11. Create a competition/prize table with Event, Participation Type, Eligibility, Registration Fee, First Prize and Second Prize.
12. Use captions, proper headers and rowspan/colspan where logically justified.
13. Use hr between major sections and br only for genuine line breaks.

Challenge: The challenge is to model one real event using different structures: rules as a collection, registration as a sequence, terminology as definitions, and schedules/prizes as structured data.

Lab Problem 5: FINAL LAB — Education & Career Fair Portal

Scenario: A college is hosting an Education & Career Fair where students can explore participating institutions and companies, compare opportunities, understand the event and find useful resources.

Requirements

1. Create a meaningful page title and h1 for the fair.
2. Write multiple paragraphs explaining purpose, audience, date, venue and benefits.
3. Create sections for About the Fair, Participating Organizations, Career Opportunities, Schedule, Eligibility, Preparation Tips, Important Information and Contact.
4. Use at least eight different taught text/formatting elements meaningfully.
5. Include a coordinator/guest message using blockquote.
6. Include a pre/code block showing a sample student profile, resume format or registration reference.
7. Use sub/sup where meaningful.
8. Add at least six relevant images such as event banner, venue, organization logos or activity photographs, with meaningful alt text.
9. Make at least three images clickable, each leading to a relevant organization, event or resource page.
10. Create useful links for Registration, Official Website, Organization Websites, Venue Information and Student Resources using taught anchor attributes.
11. Create a ul of preparation tips, an ol for the recommended student journey through the fair, and a dl defining Recruiter, Placement, Internship, Eligibility, Resume and Pre-Placement Talk.
12. Create a participant table with Organization, Sector, Opportunity Type, Eligibility, Location and Official Website, with at least eight rows and a caption.
13. Create a schedule table with Date, Time, Session, Venue and Coordinator.
14. Use proper table headings and rowspan/colspan where logically justified.
15. Use hr between major sections and br only for genuine line breaks.
16. Do not use CSS or JavaScript.

Challenge: This is the most comprehensive problem. The student must independently decide how each piece of information should be represented while demonstrating all three required areas together.

Quality Checklist

- Every problem contains text/formatting, links/images, lists and tables.
- Heading hierarchy is logical.
- Formatting elements are purposeful.
- Images have meaningful alt text and working sources.
- Links have meaningful destinations and appropriate target attributes.
- Ordered lists represent sequences; unordered lists represent collections; description lists represent terms and definitions.
- Tables represent structured data and use captions/headers correctly.
- rowspan/colspan are used only when logically necessary.
- Content is realistic and detailed.
- No CSS or JavaScript is used.